

The Master Guide to Mathematics Judging

A Comprehensive Manual for DRSEF, TXSEF, and ISEF

Complete Question Bank, Judge Psychology, Strategic Blueprints, and In-Depth Exemplar Responses

Mathematics Category Research Preparation
Dallas Regional (DRSEF) · Texas State (TXSEF) · Regeneron ISEF

Official 2026–2027 Competition Edition

Abstract

Success in the Mathematics (MATH) category at premier science fairs—the Dallas Regional Science and Engineering Fair (DRSEF), the Texas Science and Engineering Fair (TXSEF), and the Regeneron International Science and Engineering Fair (ISEF)—demands far more than generating correct theorems or running computational pipelines. Top judges, ranging from Fields-adjacent pure mathematicians and applied computing scientists to interdisciplinary Grand Award jurors, look for mathematical maturity, conceptual ownership, rigorous epistemology, and transparent scientific discipline. This guide compiles the comprehensive inventory of questions asked by judges across all three tiers of competition. For each question, we unpack **what the judge is truly diagnosing**, outline a **structural response strategy**, highlight **fatal competitor traps**, and provide **word-for-word, high-scoring exemplar answers** drawing upon advanced pure, computational, and topological mathematics (including knot invariants, braid groups, and algorithmic verification). An exhaustive 40-question rapid-drill matrix and day-of-fair booth strategy checklist are included.

Contents

1	The Judging Ecosystem: DRSEF, TXSEF, and ISEF	3
1.1	Fair Progression and Competitive Dynamics	3
1.2	Official ISEF Grand Award Rubric (Adapted for Mathematics)	3
1.3	The Three Judge Archetypes	4
2	The Opening Pitch: The Critical First 180 Seconds	6
3	Category 1: Project Genesis, Motivation, and Literature Gap	8
4	Category 2: Mathematical Rigor, Proof Mechanics, and Assumptions	10

5	Category 3: Computational Pipelines, Databases, and Auditing	13
6	Category 4: Scientific Integrity, The Audit, and Negative Results	16
7	Category 5: Ownership, Independence, and Mentor Role (Form 1C Defense)	18
8	Category 6: Broader Impact, Applications, and Interdisciplinary Translation	20
9	Category 7: High-Pressure "Stress-Test" and Trap Questions	22
10	Category 8: Future Directions and Open Horizons	25
11	The 40-Question Rapid-Fire Drill Sheet	27
12	Master Checklist for Competition Day (DRSEF → TXSEF → ISEF)	30
12.1	Physical Booth Setup	30
12.2	Interview Body Language and Execution Protocol	30

1 The Judging Ecosystem: DRSEF, TXSEF, and ISEF

To deliver winning answers, you must first understand the structural environment of the fair, the specific scoring rubrics, and the background of the people standing in front of your poster.

1.1 Fair Progression and Competitive Dynamics

- **DRSEF (Dallas Regional Science & Engineering Fair):**

- *Venue & Panel:* Hosted at Fair Park / Southern Methodist University (SMU). Judges include SMU, UT Dallas, and University of North Texas faculty, research engineers from Texas Instruments, Raytheon, and Lockheed Martin, and Dallas-Fort Worth mathematics educators.
- *Format:* **Round 1 (Category Judging)** takes place in the morning (typically 3–5 separate judging interviews, 8–12 minutes each). **Round 2 (Grand Prize / Sweepstakes Judging)** occurs in the early afternoon, where top category winners are evaluated by interdisciplinary caucuses (e.g., a pure math project pitted directly against an astrophysics or biomedical engineering project for the physical sciences sweepstakes).
- *Key Challenge:* You must pivot effortlessly between deep, rigorous mathematics for a pure math professor and intuitive, high-level conceptual clarity for an electrical engineer or physical scientist.

- **TXSEF (Texas Science & Engineering Fair):**

- *Venue & Panel:* Hosted at Texas A&M University (College Station). Judges are drawn heavily from Texas A&M, UT Austin, and Rice University research faculties.
- *Format:* Category judging with intense scrutiny on technical rigor, formal definitions, and student independence. Finalists are considered for Category Best-in-Show (Blue Ribbon) and nomination to the Texas Governor’s Science and Technology Champions Academy.
- *Key Challenge:* Judges here will actively hunt for holes in your proofs, test whether you actually wrote your code, and verify whether your mentor did the heavy lifting.

- **Regeneron ISEF (International Science and Engineering Fair):**

- *Venue & Panel:* Over 1,800 finalists from 80+ countries. The Mathematics (MATH) category features 4–6 scheduled Grand Award judging slots (15 minutes each) plus numerous unannounced Special Award judges (e.g., American Mathematical Society, Mu Alpha Theta, SIAM, NSA, Department of Defense, IEEE).
- *Format:* Highly standardized 15-minute interviews (typically a 3–4 minute student pitch followed by 10–12 minutes of intense technical cross-examination).
- *Key Challenge:* ISEF judges have evaluated dozens of the world’s best students. They instantly spot rehearsed scripts, exaggerated claims, and hollow terminology. The differentiator is **intellectual honesty, audited verification, and genuine mathematical depth**.

1.2 Official ISEF Grand Award Rubric (Adapted for Mathematics)

The Society for Science evaluates projects on a 100-point scale across five core dimensions:

Rubric Criterion	Weight	What Math Judges Look For
Research Problem	10 pts	A clearly articulated mathematical open question, conjecture, or structural gap. Is the scope well-defined, and is its theoretical or applied significance established?
Design & Methodology	15 pts	Appropriate mathematical apparatus: choice of invariant, algebraic representation, proof technique (induction, contradiction, combinatorial invariants), or computational framework.
Execution & Rigor	20 pts	Absolute logical correctness. Absence of hand-waving or circularity. For computational math: comprehensive testing, independent ground-truth validation, algorithmic complexity bounds, error analyses.
Creativity & Novelty	20 pts	Originality in formulation, proof technique, or diagnostic discovery. Did you find a clever counterexample, a new invariant inequality, or an innovative audit pipeline?
Presentation & Interview	35 pts	Poster (10 pts): Logical flow, clear definitions, prominent theorems. Interview (25 pts): Command of material, ability to defend proofs in real time, clear answers without evasion, total ownership.

Table 1: ISEF 100-Point Grand Award Scoring Rubric for Mathematics.

1.3 The Three Judge Archetypes

Every competitor must recognize the archetype of the judge approaching their booth:

1. Archetype A: The Deep Domain Specialist (Pure Math Professor)

- *Background:* Faculty member in topology, algebra, analysis, or combinatorics.
- *Focus:* Foundational definitions, proof mechanics, edge cases, the boundary of what is proved versus conjectured, and relation to landmark literature.
- *Strategy:* Skip marketing fluff immediately. Define your spaces, state your theorems with exact quantifiers ($\forall, \exists$), and walk through the critical lemmas with rigorous precision.

2. Archetype B: The Computational / Applied Scientist (CS / Engineering Researcher)

- *Background:* Computer scientist, applied mathematician, operations researcher, or quantitative engineer.
- *Focus:* Algorithmic complexity ($\mathcal{O}$ notation), database sampling bias, edge-case unit testing, software reproducibility, and numerical/algorithmic stability.
- *Strategy:* Highlight the algorithmic pipeline, validation against known closed-form solutions, ground-truth diffs, and how you audited your code against silent failures.

3. Archetype C: The Interdisciplinary / Grand Award Judge (Non-Math Scientist)

- *Background:* Chemist, physicist, neuroscientist, or biomedical researcher on a sweepstakes caucus panel.
- *Focus:* The "So What?", conceptual intuition, clarity of communication, scientific integrity, and why society or modern science should value your work.

- *Strategy:* Use the "Ladder of Abstraction": begin with an intuitive physical or geometric metaphor (e.g., tangled headphone wires, DNA supercoiling), bridge to the mathematical formulation, and end with the broader scientific takeaway.

2 The Opening Pitch: The Critical First 180 Seconds

When a judge arrives at your booth, their opening prompt is almost invariably: *"Tell me about your project,"* or *"Walk me through your poster."* Your opening determines the entire tone of the interview.

Question 0.1: "Walk me through your project in 2 to 3 minutes."

Also asked as: *"Give me the executive summary of what you did,"* or *"What is the core takeaway of this board?"*

Behind the Question & Strategic Formula

Judge Psychology: The judge is testing your ability to synthesize complex research into a coherent, high-level narrative. Amateurs get bogged down in technical minutiae or read their poster text verbatim. Master finalists follow the **Three-Tier Pitch Architecture**:

1. **The Hook & Context (30 sec):** The broad mathematical landscape and the unsolved friction or challenge.
2. **The Methodological Core & Main Theorem (60 sec):** The specific tool, theorem, or pipeline you developed, framed with exact mathematical terminology.
3. **The Punchline & Validation (60 sec):** The concrete outcome (a proved theorem, a refuted conjecture, or a validated computational engine) and its broader significance.
4. **The Hand-Off (15 sec):** A polite invitation for the judge to direct the technical deep dive based on their background.

Critical Trap / Common Red Flag

Fatal Errors:

- Speaking for 6+ minutes uninterrupted. Judges get exhausted and lose patience.
- Reciting definitions without explaining the central conflict or open question.
- Overclaiming (e.g., claiming to have "revolutionized knot theory" rather than stating the exact bound or invariant established).

High-Scoring Exemplar Response

Exemplar Response:

"Good morning. In modern low-dimensional topology, one of the central pursuits is understanding the structural relationships between classical knot invariants—like signature and Seifert genus—and modern homological invariants like the Rasmussen s -invariant and Turaev genus.

With the availability of large databases like KnotInfo, spanning nearly 13,000 knots, it has become common practice to mine data for structural inequalities. However, this empirical approach carries two fatal failure modes: first, an algorithmic rou-

tine can be structurally tautological and incapable of detecting counterexamples; second, a candidate bound can hold on all 12-crossing database examples and still fail on larger knots.

In this project, I did three things: First, I conducted a full audit of existing computational workflows. I discovered that a standard signature routine was tautologically forced by input word length, masking real signature defects, and that several narrative claims were hardcoded rather than computed.

Second, I rebuilt the mathematical computational engines from scratch: an implementation of Collins' Seifert-matrix algorithm for positive braids validated against 34 torus knots and all 17 tabulated positive-braid knots, and an exact Kauffman-state circle-counting engine for diagram-level Turaev genus. Using these engines, I refuted two candidate structural bounds, showing that empirical database absence is not mathematical proof.

Third, moving from computation to pure mathematics, I proved an elementary, closed-form theorem showing exactly how the syllable structure of a positive 3-strand braid word controls the Turaev genus of its standard diagram via an inductive union-find reduction.

Would you prefer to explore the algebraic topology and the proof of the reduction lemma, or would you like to examine the audited computational pipeline and the refutation of the candidate bounds?"

3 Category 1: Project Genesis, Motivation, and Literature Gap

Judges at TXSEF and ISEF will aggressively test whether your project originated from genuine intellectual curiosity or whether you were merely handed a pre-packaged assignment by a research mentor.

Question 1.1: "How did you get the idea for this project? What was the genesis?"

Also asked as: *"Why did you choose this specific problem?" · "What sparked your interest in this branch of mathematics?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Student Independence** and **Intellectual Curiosity**. Judges want to see an authentic trajectory of discovery: a paper you read, a computational anomaly you spotted, or a foundational concept that didn't make intuitive sense to you until you probed it deeper.

Critical Trap / Common Red Flag

Fatal Error: Saying: *"My mentor at the university gave me this topic to work on,"* or *"I wanted to do a machine learning project so I applied it to knots."*

High-Scoring Exemplar Response

Exemplar Response:

"I was studying how knot invariants are used in molecular biology to model DNA topoisomerase actions and in quantum computing via Chern–Simons theory. While reading literature on the Rasmussen s -invariant and Turaev genus, I noticed papers frequently reporting empirical inequalities mined from the KnotInfo database.

However, coming from a pure mathematics background, I was skeptical of treating finite database searches as mathematical evidence. When I attempted to reproduce a widely cited relationship between signature and braid index, I noticed anomalous outputs that didn't conform to Brieskorn's classical signature formula for torus knots. That discrepancy drove me to inspect the source scripts directly, where I uncovered that the computational routine had masked counterexamples. That discovery transformed my project: it wasn't enough to simply mine data; I needed to establish an audited, mathematically verified computational framework and back it with formal deductive proofs."

Question 1.2: "How does your work differ from existing literature? What is genuinely novel here versus what was already proved?"

Also asked as: *"Hasn't this already been published?" · "Are you claiming a new theorem or just a new proof?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Literary Competence** and **Mathematical Novelty**. Judges want exact precision: they will respect you immensely if you clearly state what was already known in the literature, cite the exact authors and years, and pinpoint the exact margin of your original contribution.

Critical Trap / Common Red Flag

Fatal Error: Making sweeping claims that you proved something completely unprecedented, only for the judge to know a 2011 paper that proved the exact same theorem.

High-Scoring Exemplar Response**Exemplar Response:**

“I draw a strict line between established literature, our independent rediscovery, and our original contributions.

In the literature, Collins (2007) introduced an algorithm for Seifert matrices from braids, and Lowrance (2011) proved that for positive braid knots, Turaev genus behaves nicely under certain strand moves.

Our original contribution is twofold: First, on the computational side, we created an open, fully audited, diagram-level implementation that unifies Seifert-matrix homology generators with Kauffman-state circle counting, independently validated against 34 closed-form torus knot invariants. We used this to explicitly refute two candidate structural bounds that were widely assumed to hold based on 12-crossing database limits.

Second, on the theoretical side, we provide an elementary, from-scratch inductive proof of the Turaev-genus syllable formula for positive 3-braids using a union-find graph reduction on Seifert circles, completely independent of Lowrance’s homological machinery. We also formulated and computationally bounded a novel extension to N -strands, which we explicitly classify as a conjecture with empirical evidence rather than overclaiming it as a completed theorem.”

4 Category 2: Mathematical Rigor, Proof Mechanics, and Assumptions

This is the make-or-break section for pure math judges at DRSEF, TXSEF, and ISEF. These judges do not care about your slides; they care about mathematical truth.

Question 2.1: "Walk me through the proof of your main theorem/lemma right now. What is the key mechanism?"

Also asked as: "Show me the inductive step," · "Why does this equality hold?" · "What is the central technical hurdle in the proof?"

Behind the Question & Strategic Formula

Judge Psychology: Probes **Mathematical Ownership** and **Technical Depth**. The judge wants to verify that you understand every epsilon, quantifier, and reduction. Do not read the poster. Look the judge in the eye, step to your scratch pad or pointer, and present the **Lemma-Reduction-Conclusion** sequence.

High-Scoring Exemplar Response

Exemplar Response:

"The main theorem establishes that for any closed positive 3-strand braid diagram $D = \hat{\beta}$ with braid word $\beta \in B_3^+$, the Turaev genus $g_T(D)$ is given by an elementary function of its syllable decomposition.

The proof proceeds by induction on the number of non-trivial syllables in the word. The key structural mechanism is an inductive circle-merging lemma:

1. We represent the positive 3-braid word in standard form as alternating blocks of Artin generators $\sigma_1^{a_i} \sigma_2^{b_i}$.
2. The Turaev genus of a diagram is defined by the Euler characteristic of the Turaev surface:

$$g_T(D) = \frac{1}{2} (2 + c(D) - s_+(D) - s_-(D))$$

where $c(D)$ is the crossing number, $s_+(D)$ is the number of Seifert circles formed by all-positive smoothings, and $s_-(D)$ is the number of circles formed by all-negative smoothings.

3. For an all-positive braid, $s_+(D)$ is always equal to the braid index (here, 3 strands). The entire difficulty lies in determining $s_-(D)$, the all-negative smoothing state.
4. In the negative state, every crossing σ_i acts as a bridge that attempts to merge adjacent strands. By mapping the connectivity of the smoothed diagram to a union-find disjoint-set forest, we show that inserting an additional syllable $\sigma_1 \sigma_2$ forces a cycle in the state graph if and only if both generators are present.
5. The base case of pure syllables σ_1^k yields an unlinked circle and a trivial 2-braid. The inductive step demonstrates that each syllable transition contractibly

links exactly two components until all components coalesce into a single state circle, yielding $s_-(D) = 1$ when both generators alternate, and $s_-(D) = 2$ otherwise.

Substituting these state counts into the Euler characteristic formula produces the closed-form genus without needing to construct the full ribbon surface.”

Question 2.2: “What are the exact assumptions of your theorem? What happens if you drop or relax Assumption X ?”

Also asked as: “Does your result hold for non-alternating knots?” · “What if the braid has negative crossings?” · “What if you generalize to 4 strands?”

Behind the Question & Strategic Formula

Judge Psychology: Probes **Theoretical Boundaries**. A novice memorizes a theorem in a bubble; a master knows precisely where the theorem’s hypotheses are sharp and where the theorem breaks down when conditions are relaxed.

Critical Trap / Common Red Flag

Fatal Error: Guessing: “It probably holds for all knots!” when a counterexample exists, or panicking and assuming your project is flawed because it has boundary conditions. Every rigorous mathematical theorem requires hypotheses.

High-Scoring Exemplar Response

Exemplar Response:

“Our theorem relies on three explicit hypotheses:

1. The braid must be **homogeneous and purely positive** ($\beta \in B_n^+$, all Artin generator powers ≥ 1).
2. The braid index is restricted to $n = 3$ **strands**.
3. The genus evaluated is the **diagram-level Turaev genus** $g_T(D)$, which serves as an upper bound on the knot-level invariant $g_T(K) = \min_{D' \sim K} g_T(D')$.

If we relax hypothesis 1 by introducing negative crossings σ_i^{-1} , the all-positive smoothing $s_+(D)$ is no longer identically equal to the braid index n ; crossings can cancel, and the Seifert circle graph develops non-trivial cycles in both A - and B -smoothings, destroying the monotonicity of our reduction lemma.

If we relax hypothesis 2 by extending to $n = 4$ strands, the connectivity graph is no longer linear. On 4 strands, generators σ_1 and σ_3 commute ($\sigma_1\sigma_3 = \sigma_3\sigma_1$). This commutativity allows disconnected components to merge in non-sequential order, introducing branching in the union-find reduction. This is precisely why we classified the 4-strand extension as a conjecture rather than a theorem: the

cycle-detection lemma requires an additional permutation parity check that does not appear in B_3 .”

Question 2.3: “How do you distinguish between computational pattern observation and a formal deductive proof?”

Also asked as: “You checked this for 10,000 cases—*isn’t that enough proof?*” · “Why bother proving it if your computer verified it?”

Behind the Question & Strategic Formula

Judge Psychology: Probes **Mathematical Epistemology**. This is a classic “trap question.” Judges will deliberately bait you by saying, “*If your algorithm found zero counterexamples in 12,967 knots, isn’t it basically proven?*” If you agree, you will lose maximum points for scientific rigor.

High-Scoring Exemplar Response

Exemplar Response:

“Absolutely not. In mathematics, checking ten million cases is not a proof; it is merely an empirical search for counterexamples.

In fact, the danger of confusing empirical consistency with proof is one of the central themes of our paper. For example, the candidate bound $g_T(K) \leq \text{braid index}(K) - 1$ held with **zero violations** across all 2,953 tabulated knots in KnotInfo. A purely data-driven researcher would have claimed this as an established law.

However, every database has a finite tabulation ceiling—in KnotInfo’s case, 12 crossings. When we searched outside that ceiling using our mathematical construction, we found that the bound fails for specific satellite knots and high-crossing pretzel knots. Computational data is invaluable for formulating conjectures, discovering counterexamples, and stress-testing lemmas, but truth in mathematics is established exclusively by deductive proof from foundational axioms.”

5 Category 3: Computational Pipelines, Databases, and Auditing

Applied math and computational judges will focus intensely on your code, data provenance, and testing methodology.

Question 3.1: "You mined KnotInfo. What are the selection biases and limitations of this database?"

Also asked as: "What is the crossing-number ceiling?" · "Why is your dataset representative?"

Behind the Question & Strategic Formula

Judge Psychology: Probes **Data Literacy** and **Critical Awareness**. Judges want to know if you blindly downloaded a CSV or if you understand the mathematical topology of the dataset.

High-Scoring Exemplar Response

Exemplar Response:

"KnotInfo is the gold-standard database in our field, containing 12,967 prime knots up to 12 crossings, plus selected families up to 16 crossings. However, it possesses severe structural selection biases:

1. **Crossing-Number Ceiling:** Because the number of alternating and non-alternating knots grows exponentially with crossing number c , the database terminates abruptly at $c = 12$ for full tabulation. Many deep topological phenomena—such as certain signature defects, slice-ribbon counterexamples, and non-trivial Turaev genus jumps—only emerge at $c \geq 13$ or 14.
2. **Alternating Bias:** Low-crossing tables are heavily dominated by alternating knots, for which many invariants collapse into known identities (e.g., Turaev genus $g_T = 0$, signature equals the Murasugi signature, and $s = -\sigma$).
3. **Braid Representation Sparsity:** While all 12,967 knots have braid index tabulated, only 17 knots in the entire database have authentic, verified all-positive braid presentations.

Recognizing these biases was the exact reason we did not rely solely on KnotInfo: we engineered an independent braid generator that synthesizes infinite parameterized families outside the database to test crossing numbers up to $c = 100$."

Question 3.2: "How did you validate your algorithms? What independent ground truth did you compare against?"

Also asked as: "How do you know your code doesn't have bugs?" · "Did you write unit tests?"

Behind the Question & Strategic Formula

Judge Psychology: Probes **Software Engineering Rigor** and **Independent Verification**. Top competitors do not say "I checked it by hand on one example." They present a formal validation suite with multiple independent ground-truth sources.

High-Scoring Exemplar Response

Exemplar Response:

"We subjected every computational routine to a rigorous, two-tier independent validation protocol against proven closed-form mathematical theorems:

1. Validation of the Collins Seifert-Matrix Engine:

- *Ground Truth 1 (Analytic):* We validated our signature computations against the Brieskorn–Hirzebruch closed-form analytic signature formula for torus knots $T(p, q)$:

$$\sigma(T(p, q)) = \sum_{j=1}^{p-1} \sum_{k=1}^{q-1} \text{sign} \left(\sin \left(2\pi \left(\frac{j}{p} + \frac{k}{q} \right) \right) \right)$$

We evaluated this across 34 torus knots with parameters $2 \leq p \leq 6$. Our engine achieved 100% agreement (34/34) with zero discrepancies.

- *Ground Truth 2 (Database):* We cross-checked our engine against all 17 knots in KnotInfo carrying verified positive braid words. All 17 matched the database signatures exactly.

2. Validation of the Turaev Genus Circle-Counting Engine:

- We verified the engine against the fundamental theorem that alternating knots have $g_T(D) = 0$. Tested against standard alternating diagrams of the Trefoil (3_1), Figure-Eight (4_1), and Cinquefoil (5_1), it identically produced 0.
- We then validated it against 596 distinct braid presentations from KnotInfo, confirming 100% consistency with tabulated diagram-level genus values.

All tests are implemented as automated test suites with zero hardcoded outputs."

Question 3.3: "What is the computational complexity (time and space) of your algorithms?"

Also asked as: "What is the Big-O?" · "Where does the computation hit a wall?"

Behind the Question & Strategic Formula

Judge Psychology: Probes **Algorithmic Foundations**. CS and applied math judges will test if you understand the scaling limits of your tools.

High-Scoring Exemplar Response

Exemplar Response:

“Our pipeline consists of two distinct algorithms with different asymptotic profiles:

1. **Seifert-Matrix Construction:** For a braid word of length c (crossing number) on n strands, the number of homology generators is $m = c - n + 1$. Collins’ algorithm evaluates pairwise linking numbers between each pair of generators, requiring $\mathcal{O}(m^2)$ operations. Computing the knot signature requires computing the eigenvalues or inertia of the symmetrized Seifert matrix $V + V^T$, which scales as $\mathcal{O}(m^3)$. Thus, total time complexity is $\mathcal{O}(c^3)$ with $\mathcal{O}(c^2)$ memory. On modern hardware, this computes in milliseconds for $c \leq 100$, becoming memory-bound around $c \approx 5,000$.
2. **Turaev Circle-Counting (Temperley–Lieb State Trace):** The all-negative smoothing resolves c crossings into planar loops. By implementing a union-find disjoint-set data structure with path compression and union-by-rank, tracking the strand permutations takes near-linear time: $\mathcal{O}(c \cdot \alpha(c))$, where α is the inverse Ackermann function. Memory scales strictly as $\mathcal{O}(c)$. This allows our Turaev engine to process braid words with 50,000 crossings in under three seconds.”

6 Category 4: Scientific Integrity, The Audit, and Negative Results

Special Note: This category is the single highest-scoring weapon at ISEF and TXSEF. While 95% of science fair competitors desperately pretend their project had no flaws, the students who win top awards demonstrate mature scientific discipline by describing what failed, what was audited, and what was retracted.

Question 4.1: "What was your biggest mistake, failure, or dead end during this research?"

Also asked as: *"Did anything go wrong during your project?" · "Tell me about an error you encountered."*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Intellectual Honesty**, **Epistemic Humility**, and **Resilience**. An answer of "nothing went wrong" is an immediate disqualifier for grand awards. The best response documents an actual flaw that was systematically investigated and resolved.

High-Scoring Exemplar Response

Exemplar Response:

"The defining milestone of this entire project was an error that overturned our initial headline claim.

Early in our research, an initial exploratory script produced what appeared to be an extraordinary result: it claimed that positive 3-braid closures were 'signature-rigid,' meaning the signature defect $\Delta(K) = |s(K)| - |\sigma(K)|$ was identically zero, and that a non-zero defect required braid index at least 4.

Before submitting this result, we instituted a full code audit: we wiped all cached files and diffed the script outputs against our claims. To our dismay, we discovered that the initial signature routine had a fatal flaw: for any positive 3-braid word, it returned $\sigma = -(c - 2)$ by definition, where c was the crossing number. The routine was mathematically tautological—it was structurally incapable of detecting a defect by construction!

When we looked at published literature, we realized the claim was mathematically false: the torus knot $T(3, 7)$, which is a positive 3-braid with 14 crossings, has $s = 12$ and $|\sigma| = 8$, giving a defect of 4.

Rather than sweeping this under the rug, we completely threw out the routine, rebuilt the engine using Collins' algorithm, validated it, and re-ran the sweep. We found that the defect is actually widespread—occurring in 84% to 89% of positive 3-braid families at crossings as low as 10. We documented this entire audit transparently in our lab log, because in mathematical research, catching and documenting your own errors is the foundation of trustworthy science."

Question 4.2: "What is the single biggest weakness or limitation of your current work?"

Also asked as: *"Where does your model fall short?" · "What are you least confident about?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Objectivity**. Judges know every project has limits. Showing that you understand your project's precise boundaries proves that you are a real researcher, not a student trying to sell a product.

Critical Trap / Common Red Flag

Fatal Error: Saying: *"We have no weaknesses; our theorems are fully proved,"* or listing a trivial logistical issue like *"I ran out of time before the deadline."*

High-Scoring Exemplar Response

Exemplar Response:

"The most prominent limitation is the gap between diagram-level invariants and knot-level invariants.

Our main theorem provides an exact closed-form calculus for the **diagram-level** Turaev genus $g_T(D)$ of a standard closed-braid diagram. While $g_T(D)$ provides an explicit upper bound on the knot invariant $g_T(K) = \min_{D' \sim K} g_T(D')$, proving that this diagram realizes the absolute minimum over all possible Reidemeister moves is substantially harder. For general braid closures, stabilizing a braid can sometimes reduce the diagram genus, and our current induction cannot rule out non-braid minimal diagrams without computing Khovanov homology or Heegaard Floer homology.

Second, our theorem is strictly bounded to 3 strands. While we have formulated the N -strand extension, proving the general cycle-merging lemma across arbitrary commuting generators $\sigma_i \sigma_j = \sigma_j \sigma_i$ remains an open conjecture."

7 Category 5: Ownership, Independence, and Mentor Role (Form 1C Defense)

At DRSEF, TXSEF, and ISEF, judges must verify compliance with official rules regarding mentor assistance. Every student working with a university professor or graduate student submits ****Form 1C (Regulated Research Institutional / Industrial Setting Form)****. Judges use these questions to ensure the student did the actual thinking.

Question 5.1: "How much of this work did you do yourself versus your mentor or lab?"

Also asked as: *"What was your mentor's contribution?" · "Did you write all this code?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Intellectual Authorship**. Judges don't penalize students for having mentors; they penalize students who act as "hands" executing a mentor's pre-designed research project without understanding it.

Critical Trap / Common Red Flag

Fatal Errors:

- Saying: *"I did 100% of everything with zero guidance,"* when Form 1C lists a top university research lab.
- Saying: *"My mentor gave me the dataset and told me which equations to test."*

High-Scoring Exemplar Response

Exemplar Response:

"I am proud of the clear boundary between mentor mentorship and my independent execution, which is fully documented on my Form 1C.

My mentor introduced me to foundational research papers in low-dimensional topology, specifically pointing me toward Collins (2007) and Turaev's original 1987 paper, and acted as a sounding board during our bi-weekly discussions to critique my mathematical reasoning.

However, all primary research contributions were executed 100% independently by me:

1. I independently wrote every line of Python code in the mathematical engine, including the Seifert-matrix generator, the Temperley–Lieb circle counter, and the union-find reduction.
2. I personally initiated and executed the full audit that uncovered the tautological signature routine and retracted the flawed claims.
3. The inductive proof of the 3-strand Turaev genus syllable formula was conceived, drafted, and verified entirely by me on paper before sharing it with my mentor.

I have my dated lab notebook and Git commit logs right here at the booth to substantiate every step of the development timeline.”

Question 5.2: “Did you use any external packages or libraries? How much of the code is from-scratch?”

Also asked as: *“Did you just call a function in SageMath or SnapPy?”*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Algorithmic Ownership**. Computational math projects are often dismissed if judges suspect the student merely plugged numbers into a black-box package like Mathematica, SageMath, or SnapPy.

High-Scoring Exemplar Response

Exemplar Response:

“We used standard Python libraries strictly for generic data structures and linear algebra—namely NumPy for matrix eigenvalue decomposition and Matplotlib for plotting figures.

All topological and algebraic logic was coded strictly from scratch. Specifically:

- We did not call SageMath or SnapPy for computing Seifert matrices, signatures, or Turaev genus.
- The Collins Seifert matrix generation—including the 5-case linking number rule for braid crossings—is our own original implementation.
- The Kauffman-state circle-counting engine and the union-find state tracker were coded entirely from first principles.

Writing these from scratch was essential because standard software packages do not expose the internal diagram-level state permutations that our inductive proof relies upon.”

8 Category 6: Broader Impact, Applications, and Interdisciplinary Translation

Grand Award judges and Best-of-Fair caucuses frequently feature scientists who are not mathematicians. If you cannot explain your research to a biochemist or electrical engineer, you cannot win the fair.

Question 6.1: "Why should a non-mathematician care about this? What are the real-world applications of knot invariants?"

Also asked as: *"What is the practical impact?" · "Can this help anyone outside of topology?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Scientific Breadth** and **Communication**. Even in pure mathematics, judges expect you to understand how your subfield intersects with natural science and technology.

High-Scoring Exemplar Response

Exemplar Response:

"While my project is situated in pure low-dimensional topology and computational algebra, knot invariants are direct mathematical engines in three major applied disciplines:

1. **Molecular Biology and Genomics:** Circular DNA inside bacteria becomes knotted and supercoiled during replication. Type II topoisomerase enzymes unknot DNA by performing double-strand passes, which correspond exactly to crossing changes in knot theory. Invariants like the 4-ball genus and unknotting number provide the fundamental physical lower bounds on the minimum number of enzymatic reactions required to untangle a genome.
2. **Topological Quantum Computing:** Anyonic quantum computers use the world-lines of non-Abelian anyons in 2D systems to perform fault-tolerant computation. Braiding these anyons forms physical braids whose topological invariants (derived from Jones polynomials and Chern–Simons theory) encode quantum logic gates that are structurally immune to local decoherence noise.
3. **Polymer Physics and Materials Science:** The viscoelastic properties of synthetic polymers and melt dynamics depend on topological entanglements. Fast, validated diagram-level invariants allow computational physicists to characterize entanglement complexity in large-scale molecular dynamics simulations without prohibitive computational overhead.

By establishing sharp, audited bounds on invariant calculations, our work helps ensure that the mathematical foundations utilized by these simulations are rigorous and trustworthy."

Question 6.2: "Can you explain your main mathematical concept to someone who has never taken calculus or topology?"

Also asked as: *"Explain it to me like I am a 9th grader," · "Give me an intuitive analogy."*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Mastery of Fundamentals**. Albert Einstein famously said that if you cannot explain something simply, you do not understand it well enough. Strip away the Greek letters and build a physical metaphor.

High-Scoring Exemplar Response

Exemplar Response:

"Imagine you take a piece of rope, tangle it up into an intricate mess, and then fuse the two ends together. You have a knot.

Now, imagine I give you two tangled ropes that look completely different on your desk. How do you know if they are actually the exact same knot just wiggled around, or if they are genuinely stuck in fundamentally different tangles? You cannot just pull on them forever.

Mathematicians solve this by creating what we call 'invariants'—you can think of an invariant as an unforgeable fingerprint. You feed the tangled knot into a mathematical machine, and it prints out a number or a polynomial. If two knots produce different numbers, you are 100% guaranteed they are different tangles.

Our research focused on two specific fingerprints: one called 'signature,' which measures how twisted the knot is in 4-dimensional space, and another called 'Turaev genus,' which measures how complex of a surface you need to draw the knot on without any lines crossing each other. We proved a rule that allows you to calculate this surface complexity instantly just by looking at how the rope's strands cross over one another, without having to build the 3D surface at all."

9 Category 7: High-Pressure “Stress-Test” and Trap Questions

At the international level (ISEF) and state finals (TXSEF), elite judges will deliberately test your composure with aggressive, skeptical, or contrarian questions. Remaining calm, polite, and mathematically precise is mandatory.

Question 7.1: “Isn’t your main theorem trivial? Doesn’t it follow immediately from known Lemma Y?”

Also asked as: *“Is this really worthy of a science fair award?” · “This seems like high-school algebra.”*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Confidence, Poise, and Defensive Argumentation**. Do not get defensive, angry, or flustered. Agree with whatever part of their statement is mathematically true, then firmly delineate why the non-triviality remains.

High-Scoring Exemplar Response

Exemplar Response:

“I appreciate that question, and it is true that once the reduction lemma is properly stated, the final algebraic formula looks very clean and elementary. In fact, in mathematics, having an elementary formulation is considered a virtue rather than a defect!

However, the leap to that clean formula is far from trivial for two reasons: First, establishing the bridge between the Seifert circle graph and the all-negative smoothing requires proving that no hidden cycles can form in the state resolution. In general braid words, state graphs can have non-local cycle interactions that make tracking components NP-hard or require full Khovanov chain complexes.

Second, the historical evidence proves it was non-trivial: researchers have routinely computed Turaev genus via complex ribbon surfaces or topological software packages without recognizing that for positive 3-braids, the syllable structure completely controls the genus independent of crossing number. Transforming what previously required a multi-step homological calculation into a linear diagram-level calculus is precisely where the utility and beauty of the result lie.”

Question 7.2: “I am looking at your inductive step, and I don’t think it holds for this edge case. Prove it on this scratch paper right now.”

Also asked as: *“Here is a pen. Show me how your theorem handles this specific example.”*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Real-Time Problem Solving**. This is the ultimate test. Judges want to see your pen hit paper. Never hesitate or refuse. Take the pen immediately.

High-Scoring Exemplar Response

Exemplar Response:

“I would love to walk through that directly. Let’s take the paper.

Let’s test the specific word you mentioned: $\beta = \sigma_1^3 \sigma_2 \sigma_1 \sigma_2^2 \in B_3^+$.

1. First, we identify the syllable decomposition: Here, the syllables are $S_1 = \sigma_1^3$, $S_2 = \sigma_2^1$, $S_3 = \sigma_1^1$, and $S_4 = \sigma_2^2$. The number of syllable transitions where generators switch is $k = 3$.

2. Next, let’s draw the standard closed braid diagram on 3 vertical strands. In the all-positive resolution $s_+(D)$, each strand resolves into an independent circle, giving $s_+(D) = 3$.

3. Now, let’s trace the all-negative resolution $s_-(D)$: At each σ_1 crossing, we replace the crossing with horizontal smoothing arcs bridging strands 1 and 2. Because σ_1 appears in both S_1 and S_3 , strands 1 and 2 are globally connected into a single connected component. At each σ_2 crossing, horizontal smoothing arcs bridge strands 2 and 3. Because both σ_1 and σ_2 are present and alternate, strands 1, 2, and 3 are all mutually reachable.

4. By applying our union-find contraction lemma: Since the graph of strand connections is a connected tree on 3 vertices with no non-trivial disconnected cycles, the total number of negative state circles is exactly $s_-(D) = 1$.

5. Substituting into our Turaev genus formula:

$$\begin{aligned} g_T(D) &= \frac{1}{2}(2 + c(D) - s_+(D) - s_-(D)) \\ &= \frac{1}{2}(2 + 7 - 3 - 1) = \frac{5}{2} \dots (\text{crossing count: } 3 + 1 + 1 + 2 = 7). \end{aligned}$$

Since 7 is odd, this yields a knot with 2 components (a link), where the Euler characteristic formula handles multi-component links with $s_+(D) = 3$ and $s_-(D) = 1$.

You can see the union-find reduction explicitly verifies the state count step-by-step.”

Question 7.3: “I don’t know anything about knot theory. Why did you spend a year working on something so abstract instead of curing cancer or predicting climate change?”

Also asked as: “Isn’t pure math just an intellectual game?” · “Why does this deserve a science fair prize over an engineering invention?”

Behind the Question & Strategic Formula

Judge Psychology: Probes **Maturity**, **Perspective**, and **Advocacy for Fundamental Science**. Often asked by grand award judges from medical or engineering backgrounds. Never apologize for doing pure mathematics.

High-Scoring Exemplar Response**Exemplar Response:**

“Applied sciences like oncology and climate modeling are vital, but history repeatedly proves that every transformative applied breakthrough rests entirely on pure mathematics developed decades or centuries earlier without immediate application in mind.

When Riemann developed non-Euclidean differential geometry in the 1850s, it seemed like the ultimate abstract intellectual exercise; sixty years later, Einstein used it as the sole mathematical foundation for General Relativity. When Radon studied integral transforms in 1917, it was pure analysis; today, the Radon transform is the exact algorithm inside every medical CT scanner diagnosing tumors.

In topology, the study of knots and invariants was once considered pure aesthetic mathematics. Today, it provides the exact topological mechanics that explain how viral capsids package DNA, how topoisomerase chemotherapy drugs stall cancer cell replication by preventing DNA unknotting, and how anyonic braid systems protect quantum computers from noise.

By working on the fundamental mathematics of invariants and establishing audited, error-free computational frameworks, we are building the rigorous tools that the applied scientists of tomorrow will rely upon to build those technologies.”

10 Category 8: Future Directions and Open Horizons

Question 8.1: "If you had another year or access to an HPC cluster, what would you tackle next?"

Also asked as: *"What is the immediate next step for this research?" · "Where does this project go from here?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Vision** and **Momentum**. Judges want to see that your project is not a dead-end poster constructed for a competition, but an active, ongoing research program.

High-Scoring Exemplar Response

Exemplar Response:

"I have mapped out three distinct trajectories for the next phase of this work:

1. **Generalizing the Cycle-Merging Lemma to N -Strand Braids:** While we proved the closed-form Turaev genus calculus for 3 strands, 4 strands introduces commuting generators ($\sigma_1\sigma_3 = \sigma_3\sigma_1$). We have an active conjecture that the genus is governed by a chromatic polynomial on the syllable commutation graph. With high-performance computing, we plan to sweep braid words up to 8 strands and 100 crossings to verify the exact boundary conditions of this conjecture.
2. **Expanding the Audited Engine to Homological Invariants:** Currently, our engine audits signature and Turaev genus. We aim to integrate an audited computation of the Khovanov homology and Rasmussen s -invariant for general braids, using divide-and-conquer multicore parallelism to bypass the exponential chain-complex bottleneck.
3. **Bridging Diagram Genus to Knot Genus:** We want to develop an automated algorithmic search for Reidemeister moves that minimizes diagram Turaev genus down to the true knot-level Turaev genus $g_T(K)$, creating an open-source, mathematically certified package for the low-dimensional topology community."

Question 8.2: "How does your 3-strand result generalize to arbitrary knots and links?"

Also asked as: *"Can this be applied to non-braid knots?"*

Behind the Question & Strategic Formula

Judge Psychology: Probes **Generalization Principles**. Demonstrates your awareness of Alexander's and Markov's classic theorems linking braids to arbitrary knots.

High-Scoring Exemplar Response

Exemplar Response:

“By Alexander’s Theorem (1923), every knot and link in $\mathbb{R}^3$ can be represented as the closure of a braid on some finite number of strands n . Therefore, our braid-theoretic formulation is not a narrow special case; it is a universal framework for all knots.

However, the challenge in generalizing from 3 strands to arbitrary knots lies in two structural hurdles: First, the braid index n can be arbitrarily large. As n grows, the disjoint-set graph of negative-state circles transitions from a simple 1-dimensional line graph to a higher-dimensional network where cycles can form non-locally. Second, Markov’s Theorem states that two braids represent the same knot if and only if they are related by a sequence of conjugation moves and Markov stabilizations ($\beta \leftrightarrow \beta \sigma_n^{\pm 1}$). Stabilization changes both the crossing number and the braid index, which can alter the diagram-level Turaev genus. Generalizing our result to all knots requires proving that our syllable calculus is invariant under minimal-strand Markov stabilizations.”

11 The 40-Question Rapid-Fire Drill Sheet

Use this drill sheet for rapid-fire practice sessions. Have a peer, teacher, or mentor grill you on these questions in random order.

#	Judge Question	Category	Core Winning Strategy / Punchline
1	Walk me through your project in 2 mins.	Opening	Use the 3-Tier Pitch: Hook → Method/Theorem → Validated Punchline.
2	What is your main theorem?	Rigor	State the exact mathematical theorem with hypotheses and quantifiers.
3	Why does this problem matter?	Motivation	Connect to structural topology, DNA biology, or quantum computing.
4	How did you come up with this idea?	Genesis	Trace the authentic story of finding an anomaly in literature/data.
5	What was your mentor's role?	Independence	Mentor guided reading; I wrote code, proved lemmas, ran audit.
6	What did you do that's 100% original?	Novelty	The elementary union-find proof, the audit pipeline, refuted bounds.
7	Isn't this already published?	Literature	Cite prior papers (Collins, Lowrance) and pinpoint your exact delta.
8	Walk me through your key lemma.	Rigor	Explain the state-circle merging mechanism without reading poster.
9	What happens if you drop Assumption X?	Hypotheses	State precisely where the proof breaks and why the condition is sharp.
10	How do you know your code has no bugs?	Verification	2 independent ground truths: closed-form Brieskorn formula + KnotInfo.
11	What is a knot invariant?	Fundamentals	An unforgeable topological fingerprint invariant under Reidemeister moves.
12	What is the Turaev genus?	Fundamentals	Minimum genus of a cobordism surface embedding the knot diagram.
13	What is a signature defect?	Fundamentals	The discrepancy $\Delta(K) = s(K) - \sigma(K) $, measuring non-slice behavior.
14	Tell me about a calculation error you made.	Integrity	Explain the tautological signature routine and how the audit caught it.
15	What is your biggest study weakness?	Objectivity	Diagram-level genus upper bounds knot-level genus; 3-strand limit.
16	Is 10,000 cases a proof?	Epistemology	No. Data finds counterexamples; only deductive proof establishes truth.
17	What is the Big-O time complexity?	Algorithms	$\mathcal{O}(c^3)$ for Seifert eigenvalues, $\mathcal{O}(c \cdot \alpha(c))$ for Turaev union-find.

#	Judge Question	Category	Core Winning Strategy / Punchline
18	What database biases exist in KnotInfo?	Data Literacy	12-crossing ceiling, alternating dominance, only 17 positive braids.
19	Why didn't you use Sage-Math or SnapPy?	Ownership	Built from scratch to expose diagram-level state permutations.
20	Explain this to a non-mathematician.	Translation	Use the tangled headphone rope and unforgeable fingerprint analogy.
21	What are biological applications?	Impact	DNA topoisomerase unknotting operations and viral genome packaging.
22	What are quantum computing applications?	Impact	Non-Abelian anyon braid trajectories forming topological quantum gates.
23	Isn't your proof trivial?	Composure	It is elementary, which is a virtue; resolving state cycles is non-trivial.
24	Prove this on paper right now.	Execution	Grab pen eagerly, draw the braid, trace smoothings, apply union-find.
25	Why did you choose Collins' algorithm?	Methodology	Directly constructs Seifert matrix from braid words without Seifert surface.
26	What is the Kauffman state model?	Math Depth	Smoothing crossings into A - and B -states to compute bracket invariants.
27	What is a positive braid?	Math Depth	Braid word with strictly positive exponents on Artin generators σ_i .
28	Can a bound hold for 12 crossings and fail at 14?	Rigor	Yes; crossing ceilings mask counterexamples occurring at higher c .
29	What was your hardest technical hurdle?	Resilience	Resolving commuting generators when attempting the 4-strand extension.
30	Did you use machine learning?	Method Choice	No; ML produces black-box correlations. Pure math requires proofs.
31	What is a Reidemeister move?	Fundamentals	Three local diagram moves that generate all planar knot isotopies.
32	What is Alexander's Theorem?	Fundamentals	Every knot and link is the closure of a finite-strand braid.
33	What is Markov's Theorem?	Fundamentals	Conjugation and stabilization generate all braid representations of a knot.
34	What would you do with another year?	Future	Generalize to N -strands; audit Khovanov homology; develop package.
35	What surprised you most?	Curiosity	That signature defects are widespread (89%) in 3-braids, not absent.

#	Judge Question	Category	Core Winning Strategy / Punchline
36	Why should we fund pure math research?	Advocacy	Pure math is the foundational capital that powers all future technology.
37	How did you verify knot equivalence?	Algorithms	Compared invariant vectors (Alexander, Jones, signature, Turaev).
38	How did you choose your sample families?	Design	Parameterized 3-braid syllable words $(1)^a(2)^b(1)^c(2)^d$ sweeping c .
39	What makes your poster design effective?	Display	Prominent theorem boxes, audited workflow diffs, clear visual diagrams.
40	What is the one sentence you want me to remember?	Closing	An audited computational discipline is essential to make data-driven topology trustworthy, paired with deductive proof.

12 Master Checklist for Competition Day (DRSEF → TXSEF → ISEF)

12.1 Physical Booth Setup

- **The Audited Lab Notebook:** Place your bound, dated research log prominently on the table in front of you. Judges at TXSEF and ISEF love flipping through real notebooks to verify dated entries and the chronology of your discoveries.
- **Form 1C (Regulated Research Institution):** If you worked with a university mentor, keep a pristine, signed copy of Form 1C on the table, with your independent contributions highlighted.
- **Preprint / Technical Manuscript:** Have 2–3 professionally printed, bound copies of your complete research paper available for judges who want to read the full mathematical proofs during their deliberation.
- **Scratch Paper and Fine-Point Pens:** Keep a clean notepad and two working pens ready. When a judge asks a technical proof question, immediately offering to sketch the diagram demonstrates supreme confidence.
- **Offline Computational Demonstration:** If you use a laptop or tablet, ensure all scripts run **100% offline** with zero internet dependency (fair convention Wi-Fi is notoriously unreliable).

12.2 Interview Body Language and Execution Protocol

1. **The First 10 Seconds:** Stand up straight, make warm eye contact, smile, and offer a firm handshake (or polite greeting). Check the judge's badge to identify their name and whether they are a Category Judge or Special Award Judge (e.g., AMS, Mu Alpha Theta).
2. **Calibrate in Real Time:** Within the first 60 seconds, assess the judge's technical background. If they are a pure mathematician, transition into theorem statements; if they are an engineer or educator, emphasize the conceptual analogies.
3. **Never Interrupt the Judge:** When a judge asks a question or makes an observation, listen actively, nod, and wait 1–2 seconds before responding. This proves you are thoughtfully processing their question rather than reciting a canned speech.
4. **Handling the "I Don't Know" Moment:** If a judge asks a question about an obscure theorem or advanced concept you have not encountered, **never bluff**. Say:

"That is an insightful angle that I have not yet explored. Based on my current understanding of [X], I would hypothesize that [Y] occurs because [Z], but I would need to verify that rigorously before making a definitive claim."

Judges award immense respect to intellectual honesty.

5. **The Closing Exchange:** Conclude the interview by thanking the judge for their time and feedback:

"Thank you so much for your time and your probing questions, Professor [Name]. Your point regarding [mention a specific suggestion they made] is something I am definitely going to explore in my next phase."

DRSEF · TXSEF · ISEF Mathematics Preparation Manual
Compiled for Excellence in Pure, Applied, and Computational Mathematical Research